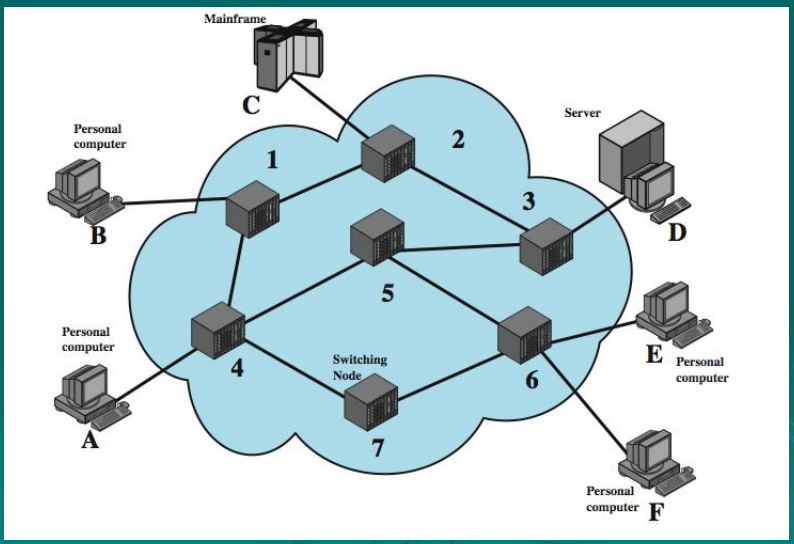
Chapter :5 Switching

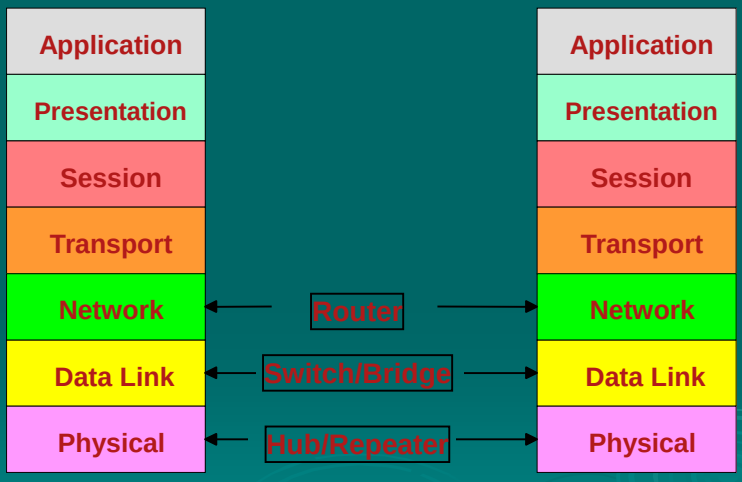
Switched Network



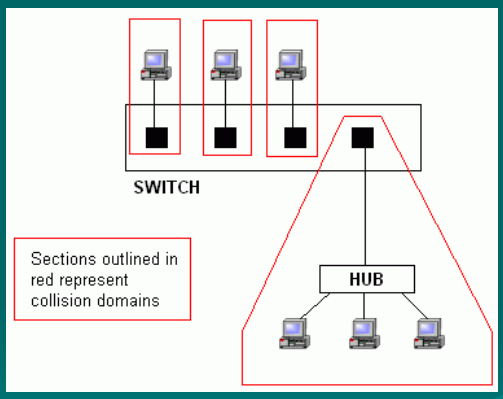
Switching

- have two different switching technologies
  - circuit switching
  - packet switching

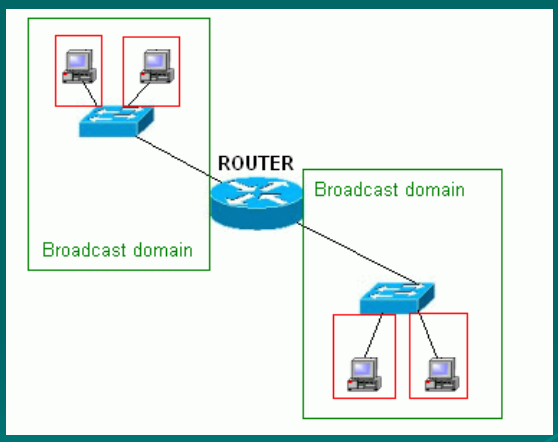
Internetworking Devices



Collision domain - example



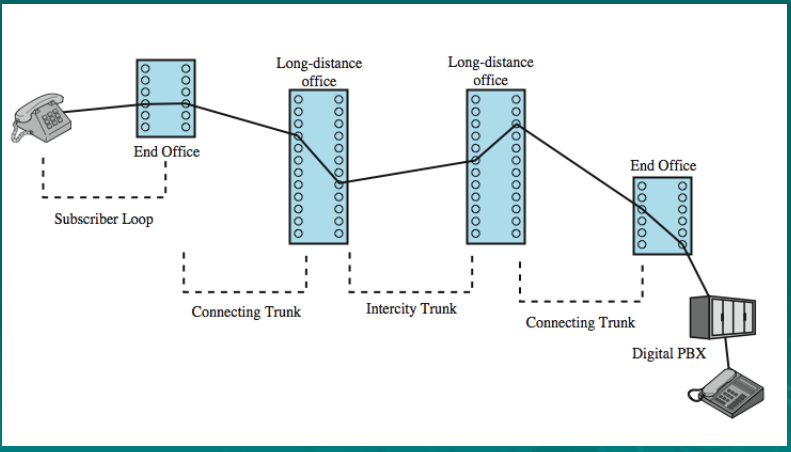
Broadcast domain – simple example



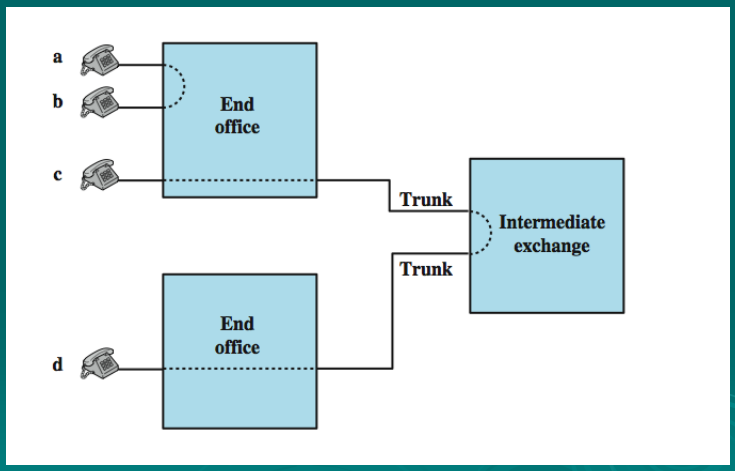
Circuit Switching

- uses a dedicated path between two stations
- has three phases
  - establish
  - transfer
  - disconnect
- inefficient
  - channel capacity dedicated for duration of connection
  - if no data, capacity wasted
- set up (connection) takes time
- once connected, transfer is transparent

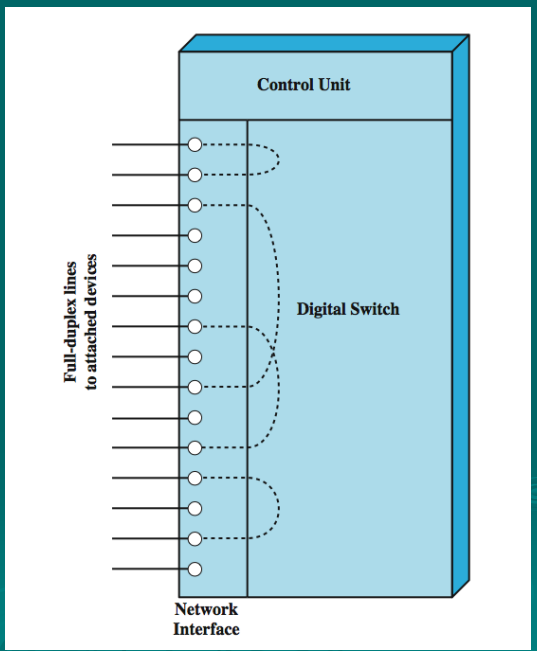
Public Circuit Switched Network



Circuit Establishment



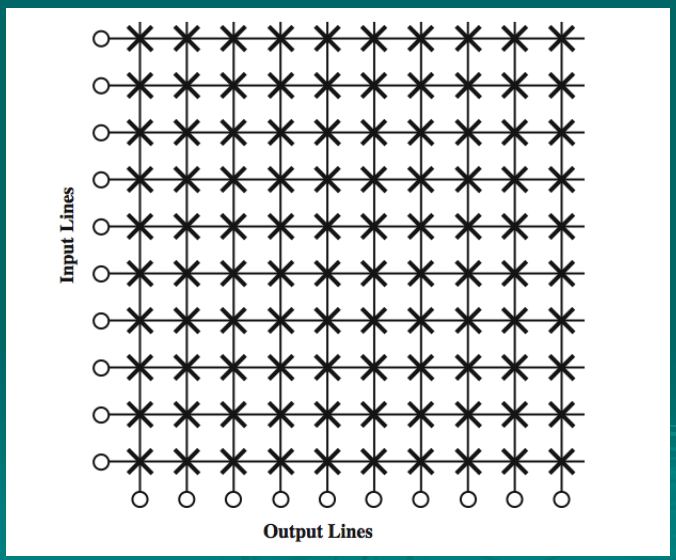
Circuit Switch Elements



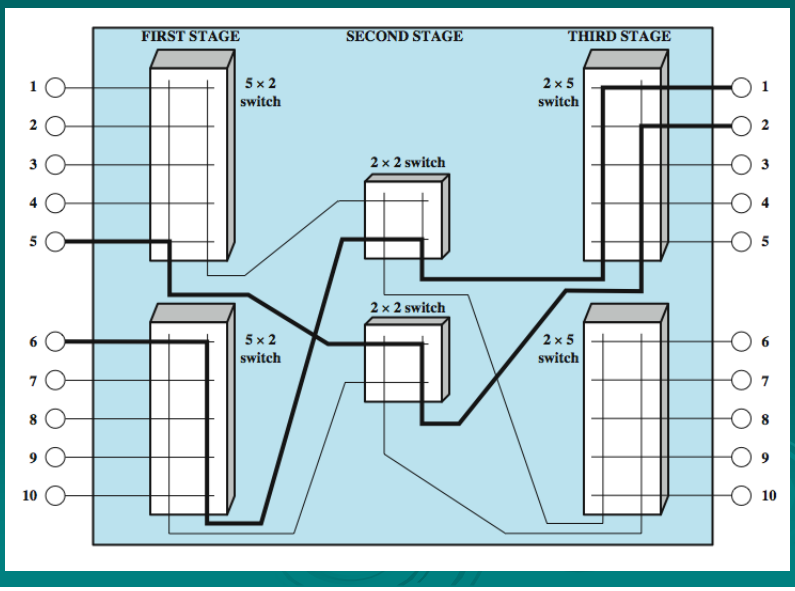
Blocking or Non-blocking

- blocking network
  - may be unable to connect stations because all paths are in use
  - used on voice systems
- non-blocking network
  - permits all stations to connect at once
  - used for some data connections

Space Division Switch



3 Stage Space Division Switch

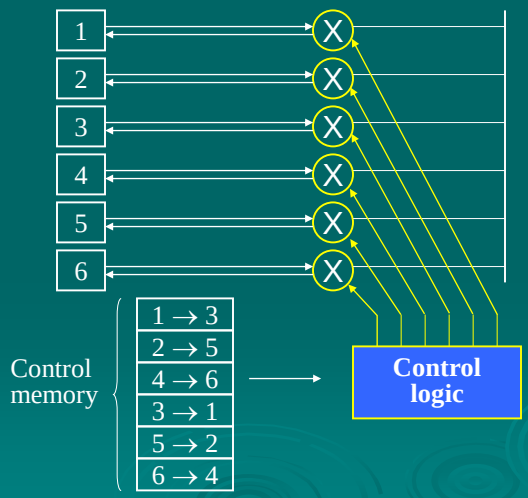


Time Division Switching

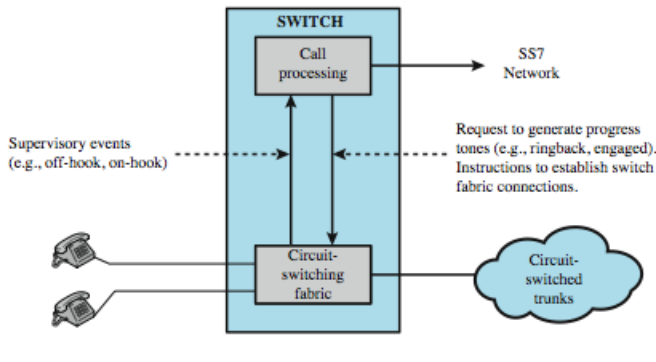
- modern digital systems use intelligent control of space & time division elements
- use digital time division techniques to set up and maintain virtual circuits
- partition low speed bit stream into pieces that share higher speed stream
- individual pieces manipulated by control logic to flow from input to output

Time-division Switching (cont)

- **Control of a TDM bus switch**



Traditional Circuit Switching

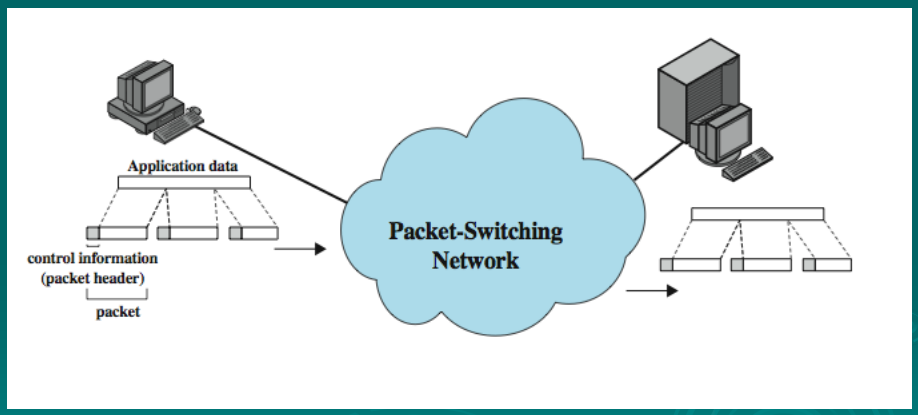


(a) Traditional circuit switching

Packet Switching

- circuit switching was designed for voice
- packet switching was designed for data
- transmitted in small packets
- packets contains user data and control info
  - user data may be part of a larger message
  - control info includes routing (addressing) info
- packets are received, stored briefly (buffered) and past on to the next node

Packet Switching



Advantages

- line efficiency
  - single link shared by many packets over time
  - packets queued and transmitted as fast as possible
- data rate conversion
  - stations connects to local node at own speed
  - nodes buffer data if required to equalize rates
- packets accepted even when network is busy
- priorities can be used

Switching Techniques

- station breaks long message into packets
- packets sent one at a time to the network
- packets can be handled in two ways
  - datagram
  - virtual circuit

Virtual Circuits vs. Datagram

- virtual circuits
  - network can provide sequencing and error control
  - packets are forwarded more quickly
  - less reliable
- datagram
  - no call setup phase
  - more flexible
  - more reliable